



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

SPECTROSCOPY OF ASTROPHYSICAL PLASMAS

Fritz Ali Agildere

June 9, 2026



Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.



Left to Right, Top to Bottom: Abell 85, Abell 426 (Perseus), Abell 1644, Abell 1656 (Coma), Abell 2029, and Abell 3158.

Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

Motivation

Motivation

- Natural Central SMBH Rapid Feeding Consequence

Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Potentially Galaxy Merger Triggered

Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Potentially Galaxy Merger Triggered
- Vertical Gas & Dust Expansion Intercepts Sightlines

Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Potentially Galaxy Merger Triggered
- Vertical Gas & Dust Expansion Intercepts Sightlines
- Fuel Region / Galactic Disk Kinematic Decoupling

Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Potentially Galaxy Merger Triggered
- Vertical Gas & Dust Expansion Intercepts Sightlines
- Fuel Region / Galactic Disk Kinematic Decoupling
- Spatial SF & AGN Luminosity Correlation

Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Potentially Galaxy Merger Triggered
- Vertical Gas & Dust Expansion Intercepts Sightlines
- Fuel Region / Galactic Disk Kinematic Decoupling
- Spatial SF & AGN Luminosity Correlation
 - Galactic Scales $R \sim 10$ kpc Weak

Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Potentially Galaxy Merger Triggered
- Vertical Gas & Dust Expansion Intercepts Sightlines
- Fuel Region / Galactic Disk Kinematic Decoupling
- Spatial SF & AGN Luminosity Correlation
 - Galactic Scales $R \sim 10 \text{ kpc}$ Weak
 - Compact Nucleus $R \sim 100 \text{ pc}$ Strong

Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Potentially Galaxy Merger Triggered
- Vertical Gas & Dust Expansion Intercepts Sightlines
- Fuel Region / Galactic Disk Kinematic Decoupling
- Spatial SF & AGN Luminosity Correlation
 - Galactic Scales $R \sim 10 \text{ kpc}$ Weak
 - Compact Nucleus $R \sim 100 \text{ pc}$ Strong
- Dynamically Evolving Obscuring Starforming Structure



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

Structure

Structure

- Radiatively Supported Large Scale Heights (Star Formation)

Structure

- Radiatively Supported Large Scale Heights (Star Formation)
 - Intense Stellar Radiation Pressure

Structure

- Radiatively Supported Large Scale Heights (Star Formation)
 - Intense Stellar Radiation Pressure
 - Supernovae Kinetic Feedback

Structure

- Radiatively Supported Large Scale Heights (Star Formation)
 - Intense Stellar Radiation Pressure
 - Supernovae Kinetic Feedback
- Distinct Peripheral $R > 10 \text{ pc}$ Regions (Compton Thin)

Structure

- Radiatively Supported Large Scale Heights (Star Formation)
 - Intense Stellar Radiation Pressure
 - Supernovae Kinetic Feedback
- Distinct Peripheral $R > 10 \text{ pc}$ Regions (Compton Thin)
- Highly Localized $R < 1 \text{ pc}$ Obscuration (Compton Thick)

Structure

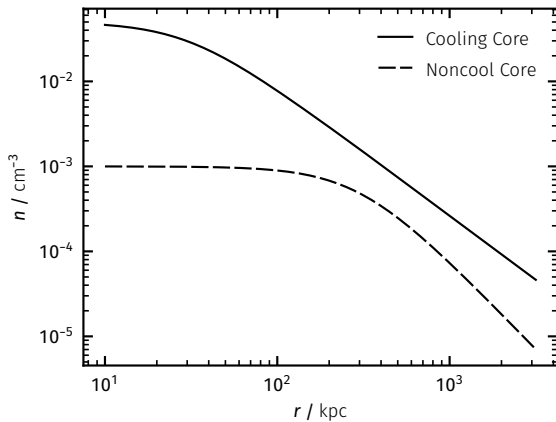
- Radiatively Supported Large Scale Heights (Star Formation)
 - Intense Stellar Radiation Pressure
 - Supernovae Kinetic Feedback
- Distinct Peripheral $R > 10 \text{ pc}$ Regions (Compton Thin)
- Highly Localized $R < 1 \text{ pc}$ Obscuration (Compton Thick)
- Embedded Stellar Remnant Thermal Effects

Structure

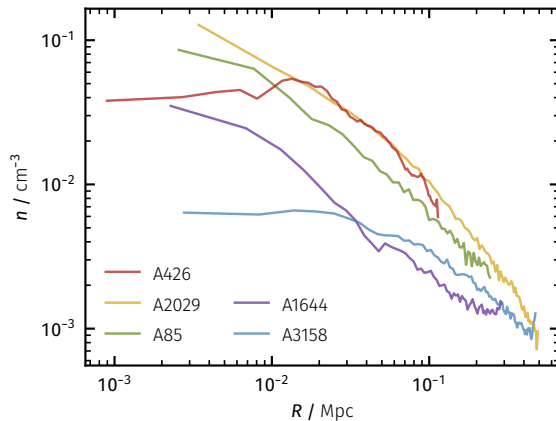
- Radiatively Supported Large Scale Heights (Star Formation)
 - Intense Stellar Radiation Pressure
 - Supernovae Kinetic Feedback
- Distinct Peripheral $R > 10 \text{ pc}$ Regions (Compton Thin)
- Highly Localized $R < 1 \text{ pc}$ Obscuration (Compton Thick)
- Embedded Stellar Remnant Thermal Effects
- Statistically Consistent Viewing Angle Sampled Distributions

Density

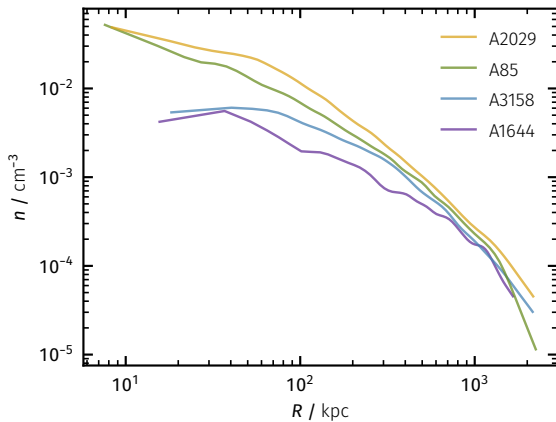
Density



Density

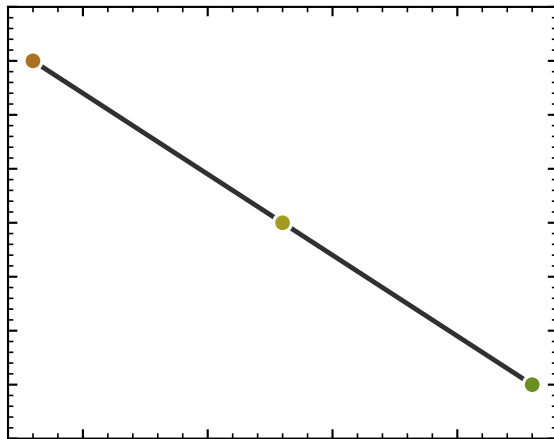
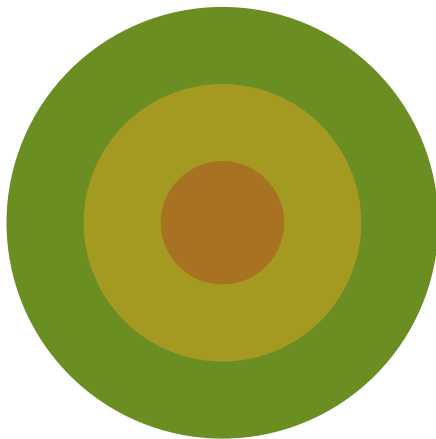


Density

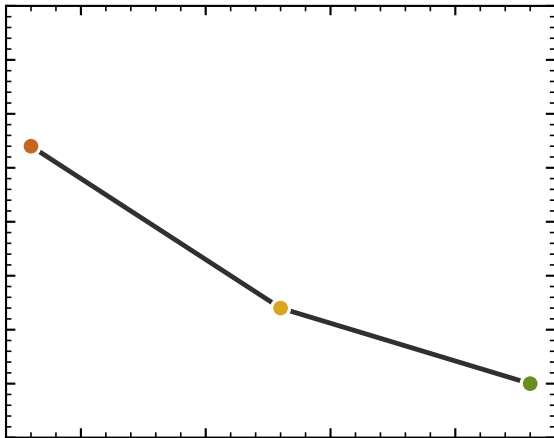


Deprojection

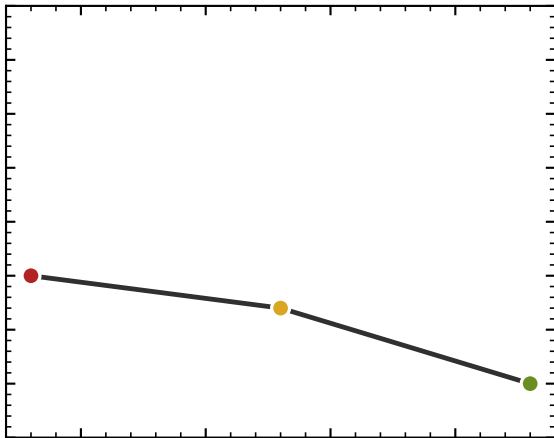
Deprojection



Deprojection



Deprojection





MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

Outlook

Outlook

- Observational Static Torus / Puffy Compact Starburst Disk Separation

Outlook

- Observational Static Torus / Puffy Compact Starburst Disk Separation
- Evolutionary Stage / Static Mechanism Distinction

Outlook

- Observational Static Torus / Puffy Compact Starburst Disk Separation
- Evolutionary Stage / Static Mechanism Distinction
- Refined Simulation Feedback Interactions (FIRE & STARFORGE)

Outlook

- Observational Static Torus / Puffy Compact Starburst Disk Separation
- Evolutionary Stage / Static Mechanism Distinction
- Refined Simulation Feedback Interactions (FIRE & STARFORGE)
- CO SLED Molecular Gas Tracers (ALMA)

Outlook

- Observational Static Torus / Puffy Compact Starburst Disk Separation
- Evolutionary Stage / Static Mechanism Distinction
- Refined Simulation Feedback Interactions (FIRE & STARFORGE)
- CO SLED Molecular Gas Tracers (ALMA)
- IR Starforming & Disk Signatures (JWST)

Thank You!

- R. C. Hickox & D. M. Alexander, *Obscured Active Galactic Nuclei* (2018) ARA&A Volume 56
- P. F. Hopkins et. al., *Stellar and Quasar Feedback in Concert: Effects on AGN Accretion, Obscuration and Outflows* (2016) MNRAS Volume 458 Issue 1

Resources on GitHub: [here](#)